

Bronchiolitis Revision Booklet

A clearer, calmer guide for student nurses. This version keeps the clinical detail you need, but explains it in a more practical way: what bronchiolitis is, why infants deteriorate, how to spot red flags, and what your nursing priorities are on the ward.

1

core case scenario threaded through the booklet

A–E

structured assessment page you can actually use

8

student-friendly sections with key takeaways

NICE

management thresholds checked against UK guidance

SCENARIO · ELLIE, 10 MONTHS

Start with the child, not the diagnosis.

Ellie has had 3 days of cough, coryzal symptoms, and worsening feeding. She now has **RR 62/min**, **SpO₂ 90% on air**, intercostal and subcostal recession, grunting, and weight loss. Her mum Sara is distressed and has older children at home.

RR 62/min

SpO₂ 90% on air

Intercostal recession

Subcostal recession

Grunting

Difficulty feeding

Weight loss

WHAT BRONCHIOLITIS ACTUALLY IS

Keep the definition simple.

- Bronchiolitis is a **viral lower respiratory tract infection** affecting the bronchioles.
- It is most commonly caused by **RSV**, although other viruses can cause it too.
- It usually starts like a cold, then progresses to **increased work of breathing**, poor feeding, and hypoxaemia in more severe cases.
- Treatment is mainly **supportive**: oxygen if needed, fluids, feeding support, and escalation when the child is tiring.

WHY INFANTS GET SICKER FASTER

The bedside explanation.

- Infants have **smaller bronchioles**, so swelling and mucus block airflow more easily.
- They have **less respiratory reserve**, so they tire more quickly.
- Breathing and feeding compete with each other, so **poor feeding is a clinical sign**, not just a comfort issue.
- Young infants are largely **nose breathers**, so upper airway secretions can worsen both feeding and breathing.

WHAT YOU SHOULD BE ABLE TO DO AFTER THIS BOOKLET

Recognise severity

Spot the red flags that tell you the child needs urgent escalation.

Explain the physiology

Link mucus, oedema, airway narrowing, and low oxygen saturations.

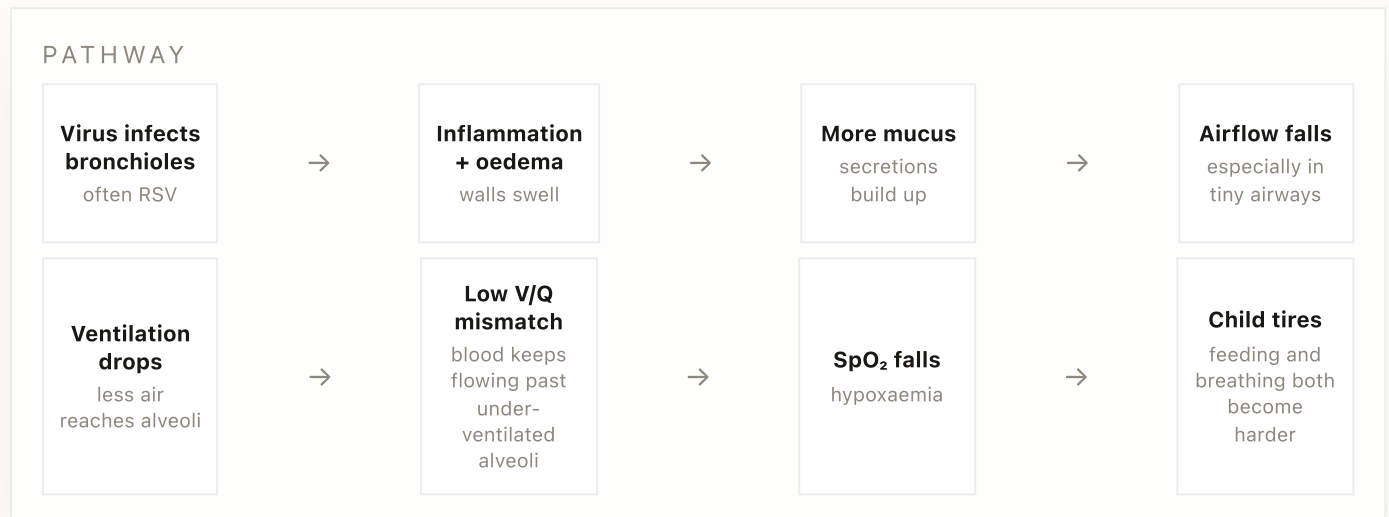
Apply A–E

Move from structured assessment to practical nursing action.

PATHOPHYSIOLOGY MADE CLEARER

What is happening in Ellie's lungs?

Start with the airflow problem. A child with bronchiolitis is not struggling because the lungs have suddenly “forgotten how to breathe”. The problem is that inflamed, mucus-filled bronchioles make it much harder to move air where it needs to go.



THREE LAWS WORTH KNOWING

Fick's law

Gas exchange works best with a large surface area, a steep pressure gradient, and a thin membrane. Inflammation and oedema make diffusion less efficient.

Boyle's law

When thoracic volume rises, pressure falls and air moves in. When volume falls, pressure rises and air moves out.

Poiseuille's law

Airway resistance rises sharply when radius falls. In a baby's already narrow bronchioles, even small swelling makes breathing much harder.

TRANSLATE THE PHYSIOLOGY AT THE BEDSIDE

- **Tachypnoea** means the child is trying to maintain minute ventilation.
- **Recession** means the child is working hard to create negative pressure and pull air in.
- **Grunting** is a red flag because the child is creating intrinsic PEEP to keep alveoli open.
- **Poor feeding** often means work of breathing is so high that the child cannot safely coordinate suck, swallow, and breathe.
- **Lower oxygen saturations** reflect under-ventilated alveoli, not just “a bit of wheeze”.

Clinical translation for Ellie

Ellie's RR of 62, recession, grunting, feeding difficulty, and SpO₂ 90% on air all fit the same story: her bronchioles are inflamed and narrowed, air movement is poor, and the effort needed to maintain oxygenation is becoming unsustainable.

ASSESSMENT

What signs matter most in infants?

Student answers are stronger when they move beyond listing signs and explain what those signs mean. Use the child's work of breathing, oxygenation, hydration, and level of alertness to judge severity.

SIGN	WHAT IT SUGGESTS	WHY IT MATTERS TO NURSING CARE
Tachypnoea	The child is compensating for reduced ventilation.	Repeat observations, watch for fatigue, and trend PEWS rather than relying on one number.
Intercostal or subcostal recession	Increased work of breathing and strong negative intrathoracic pressure.	Escalate if worsening; this is not a "minor" sign in infants.
Grunting	Intrinsic PEEP is being generated to keep alveoli open.	Severe respiratory distress; needs urgent review and escalation.
Poor feeding or weight loss	Breathing and feeding are competing; hydration and nutrition are at risk.	Think fluid balance, NG/OG support, and family-centred explanation early.
Apnoea, listlessness, or reduced effort	Possible impending respiratory failure.	Emergency escalation. Reduced effort can be more dangerous than noisy breathing.
Low oxygen saturation	Hypoxaemia from poor ventilation and low V/Q mismatch.	Follow age-specific thresholds and local escalation policy.

Red flags needing urgent escalation

- Apnoea
- Looks seriously unwell
- Severe respiratory distress: grunting, marked chest recession, or RR >70/min
- Central cyanosis
- Signs of exhaustion or recurrent apnoea

NICE thresholds worth remembering

- Consider hospital referral if RR >60/min, intake is only 50% to 75% of usual, there is clinical dehydration, or persistent SpO₂ is <92% in air.
- Admit if persistent SpO₂ is <90% in children aged 6 weeks and over, or <92% in babies under 6 weeks or children with underlying conditions.

WHAT THIS MEANS FOR ELLIE

Ellie is already beyond "watch and wait".

She has grunting, significant work of breathing, feeding difficulty, and a saturation of 90% on air. That combination is enough to justify urgent senior review, oxygen as prescribed, close monitoring, and active planning for hydration and escalation if she tires further.

STRUCTURED ASSESSMENT

A–E keeps your thinking organised.

A strong paediatric assessment is systematic. You are not trying to sound clever; you are trying to notice deterioration early and act on what you find.

A • Airway

If Ellie is crying or vocalising, the airway is patent. Check whether upper airway secretions are making breathing or feeding harder. Position for comfort, minimise agitation, and only suction if secretions are clearly contributing.

B • Breathing

RR 62/min, recession, grunting, and SpO₂ 90% on air make breathing your biggest concern. Apply oxygen as prescribed, start continuous saturation monitoring, repeat full respiratory observations, and escalate.

C • Circulation

Check HR, CRT, colour, temperature of hands and feet, urine output, and hydration status. Weight loss and poor feeding matter because they increase the risk of dehydration and fatigue.

D • Disability

Is Ellie alert, settling, irritable, or becoming listless? Reduced responsiveness can signal worsening hypoxaemia or exhaustion. Consider blood glucose if intake has been poor.

E • Exposure

Check temperature, rash, hydration, weight, and the overall picture. Reassess the child after each intervention rather than treating the A–E as a one-off task.

IMMEDIATE NURSING ACTIONS

- 1 Escalate early to the paediatric team or according to local PEWS policy.
- 2 Give oxygen if saturation meets the threshold for supplementation.
- 3 Trend work of breathing, not just the saturation number.
- 4 Start strict fluid balance and think early about feeding support.
- 5 Document clearly and reassess from A after any intervention.

Parent concerns matter

Sara knows Ellie's normal behaviour, breathing pattern, and feeding routine. If a parent is telling you the child is "not right", that is clinically relevant information. Use the trust's rapid review or deterioration pathway, including Martha's Rule processes where available locally.

COMMON STUDENT MISTAKE

Do not treat A–E like a script with no action attached.

A better answer sounds like this: "On breathing, Ellie has RR 62/min, recession, grunting, and SpO₂ 90% on air, suggesting significant respiratory distress and hypoxaemia. I would escalate urgently, apply oxygen as prescribed, increase observation frequency, and reassess after intervention."

MANAGEMENT

What your nursing priorities are on the ward

Bronchiolitis management is mainly supportive. The nurse's role is central because the child's condition can change quickly, and the family needs clear, steady explanation while that is happening.

DO

- Give oxygen if saturation is persistently below the relevant NICE threshold.
- Consider upper airway suctioning if secretions are worsening respiratory distress or feeding.
- Use NG or OG fluids if the child cannot take enough orally.
- Use IV isotonic fluids if NG/OG is not tolerated or there is impending respiratory failure.
- Escalate early if the child is tiring, becoming less responsive, or needing increasing oxygen support.
- Support positioning, cluster care where possible, and reduce unnecessary distress.

DO NOT ROUTINELY USE

- Antibiotics
- Salbutamol
- Nebulised adrenaline
- Hypertonic saline
- Systemic or inhaled corticosteroids
- Chest physiotherapy in children without relevant comorbidities
- Blood gas testing unless there is severe worsening distress or suspected impending respiratory failure

Feeding and hydration are not side issues

For infants, feeding is part of respiratory assessment. If the child is taking much less than normal, desaturating during feeds, or losing weight, you need to think ahead about fluids, NG/OG support, and how to explain that plan to the parent.

Little and often if safe

Strict input/output

Daily weight if needed

NG/OG if oral intake inadequate

When to think bigger than standard ward care

If the child shows signs of exhaustion, recurrent apnoea, or is failing to maintain adequate saturation despite oxygen, you need urgent senior review. NICE advises considering CPAP in babies and children with impending respiratory failure.

NURSING PRIORITIES IN THE FIRST HOUR

Observe

Trend RR, work of breathing, SpO₂, PEWS, responsiveness, and feeding tolerance.

Support

Give oxygen or fluids as prescribed, position well, and minimise unnecessary handling.

Communicate

Tell the senior team what is changing, and keep the parent updated in plain language.

HOLISTIC NURSING CARE

The child is unwell, but the family is under pressure too.

Ellie is 10 months old. Hospital is noisy, unfamiliar, and separating. Her mum Sara is worried and also trying to manage siblings and home life. Nursing care needs to address both the physiology and the experience.

SUPPORT THE CHILD

- Keep care calm and predictable where possible.
- Use parent presence, comfort positioning, and familiar objects to reduce distress.
- Cluster tasks so the child gets proper rest between observations and interventions.
- Remember that crying increases oxygen demand and can worsen apparent work of breathing.

SUPPORT THE PARENT

- Explain what bronchiolitis is in plain language and say what you are watching for.
- Validate distress rather than treating it as "overreacting".
- Include Sara in comfort measures and feeding discussions if she wants to be involved.
- Ask about practical pressures: transport, childcare for siblings, overnight stay, and confidence about discharge.

WHO MIGHT BE INVOLVED

Staff nurse

Coordinates observations, oxygen, fluid balance, feeds, escalation, and family communication.

Paediatric team

Confirms diagnosis, prescribes treatment, and decides on escalation or higher-level care.

Play specialist

Helps reduce distress and make the environment more manageable for the child.

Dietitian or feeding support

Useful if intake is poor, weight loss is significant, or feeding issues persist.

Social support

Important where transport, finances, siblings, or home pressures affect care.

GP / health visitor

Key after discharge for recovery, feeding, growth, and reassurance at home.

Discharge checklist

- Clinically stable
- Adequate oral intake
- Maintains oxygen saturation in air for 4 hours including sleep
- Parent understands red flags and when to seek help

Saturation targets before discharge

NICE advises maintaining sats in air at **over 90%** for children aged 6 weeks and over, or **over 92%** for babies under 6 weeks or children with underlying health conditions, for 4 hours including a period of sleep.

OTHER COMMON RESPIRATORY CONDITIONS

What else should be on your radar?

A breathless child is not automatically “just bronchiolitis”. For student nurses, the useful skill is pattern recognition: what kind of noise is the child making, where is the airway problem likely to be, and what needs escalating first?

CONDITION SPOTTERS

Bronchiolitis

Usually infants. Coryza, cough, poor feeding, recession, wheeze or crackles. Lower small-airway problem. Supportive care is key.

Croup

Barking cough, hoarse voice, inspiratory stridor, often worse at night. Think upper airway narrowing. Keep the child calm and upright.

Viral wheeze / asthma

Expiratory wheeze, breathlessness, cough, prolonged expiration. Often triggered by viral illness, exercise, or allergens. Use the child's action plan and local pathway.

Pneumonia

Fever, cough, increased work of breathing, and sometimes focal crackles. High fever or persistently focal chest findings make this differential more likely.

FAST PATTERN RECOGNITION

- **Bark + stridor** points you towards croup.
- **Expiratory wheeze + prolonged expiration** fits wheeze or asthma more than bronchiolitis.
- **Infant + coryza + poor feeding + recession** keeps bronchiolitis high on the list.
- **High fever + focal crackles** makes pneumonia more likely and is specifically highlighted in NICE bronchiolitis guidance as a differential to consider.

Escalate before you over-label

If the child is blue, apnoeic, increasingly drowsy, drooling, or failing to maintain oxygenation, the immediate priority is escalation and airway/breathing support, not arguing about the perfect diagnosis.

Useful nursing distinction

Not every preschool child with wheeze has asthma. Viral-induced wheeze is common in younger children and often occurs only during viral illnesses, which is why history and pattern matter.

SAFE HIGH-LEVEL FIRST THOUGHTS

Croup

Keep calm, sit upright, avoid steam, and recognise when stridor at rest or severe breathing difficulty needs urgent review.

Wheeze / asthma

Think inhaler via spacer, response to treatment, and follow the child's action plan or local acute wheeze pathway.

Pneumonia

Think infection, hydration, sepsis screening if very unwell, and antibiotic treatment when clinically indicated.

COMMUNICATION AND QUICK REVISION

Finish with a handover you could actually give.

Students often know the content but lose marks because their communication sounds vague. SBAR helps you say the important things clearly and ask for what the child needs next.

S

Situation

"I'm calling about Ellie, a 10-month-old with bronchiolitis. I'm concerned because her work of breathing is high and her SpO₂ is 90% on air."

B

Background

"She has had 3 days of cough, coryzal symptoms, worsening feeding, and weight loss. Mum is at the bedside and is very worried."

A

Assessment

"RR is 62/min, she has intercostal and subcostal recession, grunting, and looks fatigued. Her feeding has worsened."

R

Recommendation

"Please review her urgently. I need support with oxygen management, escalation, and a plan for feeding and hydration."

EXAM QUICK WINS

- Explain the sign, then link it to what you would do.
- Say "supportive management" and then name what that means.
- Remember that poor feeding and parent concern are clinically important.
- Use A-E to organise your answer if the question feels messy.
- State when bronchiolitis treatments are **not** recommended.

COMMON MISTAKES

- Calling bronchiolitis a bacterial illness and defaulting to antibiotics.
- Focusing only on wheeze and ignoring feeding, hydration, and fatigue.
- Listing observations with no interpretation.
- Writing about family-centred care without linking it to nursing action.
- Forgetting age-specific saturation thresholds.

FINAL TAKEAWAYS

End with the points examiners actually want.

The safest bronchiolitis answers focus on observation, supportive care, escalation, and communication. Keep these takeaways in your head and your answer will sound more clinical and more nursing-focused.

SOURCES CHECKED FOR THIS VERSION

NICE NG9, *Bronchiolitis in children: diagnosis and management* (published 2015, updated 2021): referral, admission, oxygen, suctioning, fluids, discharge, and treatments not recommended.

NICE NG250, *Pneumonia: diagnosis and management* (published 2 September 2025): community-acquired pneumonia in babies and children, including updated antibiotic guidance.

NHS.UK, *Croup*: hallmark symptoms, home care advice, and emergency red flags.

NHS.UK, *Asthma*: symptom overview and attack awareness. Acute wheeze or asthma management should always follow the child's action plan and local pathway.

NHS England, *Martha's Rule information for patients, families and carers*: rapid review concept and family escalation context.

This booklet is for revision and does not replace local policy, PEWS guidance, medicines management, or trust escalation procedures.

One sentence summary

Bronchiolitis is mainly a supportive-care illness: recognise worsening work of breathing early, protect oxygenation and hydration, escalate before the child tires, and keep the family informed all the way through.

- Recognise
- Support
- Escalate
- Reassure

PARENT-FACING LANGUAGE

Strong nursing answers sound calm, specific, and honest. Use clear family language as well as clinical terms.

Explain simply "This is usually a viral breathing illness, so we're closely watching breathing, oxygen levels, and feeding."	Name the worry "What concerns us is that the breathing looks harder work and they're not drinking as well."	Say what you are doing "We're repeating observations, supporting oxygen and fluids if needed, and getting senior review early."	Safety-net clearly "If breathing gets noisier, your baby gets more sleepy, or feeding drops further, tell us straight away."
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